

JEAN SALAC, PHD

Assistant Professor of Computer Science | Computing Education Research & Human-Centered Computing

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EDUCATION

- 2021 PhD | Computer Science | **University of Chicago** | National Science Foundation Graduate Research Fellow
- 2020 MS | Computer Science | **University of Chicago**
- 2017 BS | Computer Science | **University of Virginia** | High Distinction

PROFESSIONAL APPOINTMENTS

- 2024-Present Assistant Professor of Computer Science, Carleton College
- 2022-24 Postdoctoral Researcher & Computing Innovations Fellow, University of Washington
 - Mentor : Prof. Amy J. Ko
 - Researched critical computing education for youth, namely how youth may learn to engage in the moral and ethical complexities of the world with computing, and how educators can facilitate this engagement
- 2021 AI/ML Education Researcher, Apple
 - Mentor : Prof. R. Benjamin Shapiro
 - Explored opportunities to introduce data science practices to K-12 students using Swift Playgrounds & Charts
 - Incorporated the latest findings in computing education research into Apple's education strategy
- 2017-21 Graduate Researcher, University of Chicago
 - Mentor : Prof. Diana Franklin
 - Applied human factors, statistical, and machine learning techniques to analyze student data
 - Built a tool for automatic generation of personalized written assessments that incorporates student code, and a tool that scraped features of student code for static analysis
- 2016-17 Undergraduate Researcher, University of Virginia
 - Mentors : Prof. Luther Tychonievich & Prof. Rider Foley
 - Researched problematic demographic categories used in STEM higher education
 - Investigated the impact of informal computing education on low-income female youth of color
- 2016 Computer Science Education Fellow, National Science Foundation
 - Mentor : Dr. Jan Cuny
 - Analyzed prior NSF CS education and broadening participation initiatives
 - Researched social innovation best practices to help shape President Obama's CS for All initiative

HONORS, AWARDS, AND PROMINENT APPOINTMENTS

- 2026-2027 **Guest Editor**, Transactions on Computing Education (TOCE) Special Issue on "Situated and Critical Effects of Generative AI in Computing Education : Identity, Belonging, and the Social Conditions of Learning"
- 2025 **Distinguished Paper** | ACM Transactions on Computing Education (TOCE)
- 2024 **Best Paper Honorable Mention** | ACM Conference on International Computing Education Research (ICER)
- 2023 **Best Paper** | ACM Conference on International Computing Education Research (ICER)
- 2022-23 Advisory Board | CodePath
- 2022-24 **Computing Innovations Fellow** | National Science Foundation & Computing Research Association
- 2021 William Rainey Harper Dissertation Award | University of Chicago's Highest Honor for Graduate Students
- 2020-21 Graduate Fellow | Computing Research Association - Education
- 2020 EECS Rising Star | Rising Stars Academic Career Workshop for Women
- 2020 **Best Paper** | ACM Conference on International Computing Education Research (ICER)
- 2020 Online Organizing Committee Member | IEEE RESPECT Conference for Equity and Sustained Participation
- 2019 **Graduate Research Fellowship**, | National Science Foundation
- 2019 Bridge Builder Leadership Award, Computer Science Department Teaching Award, Physical Science Division Teaching Award Nomination | University of Chicago
- 2018 Graduate Student Leadership Award | University of Chicago
- 2018 **Best Paper Honorable Mention** | ACM CHI Conference on Human Factors in Computing Systems (CHI)
- 2017 Rader Award for Undergraduate Research | University of Virginia
- 2016 AAPI Young Leader | White House Initiative for Asian-Americans and Pacific Islanders (WHIAAPI)

PUBLICATIONS

In computing, conference papers undergo a highly selective, multi-stage peer review process and are considered on par with journal publications. In my research sub-fields, it is customary to list the supervisor last in the author order (e.g. professor supervising the research team). In cases where there is no supervising author, authors are listed in order of contribution. Approximate conference acceptance rates: ICLS, IDC, ITiCSE, SIGCSE ~25-30%; CHI, ICER ~20-25%. † denotes mentored students.

- 2025 **Jean Salac**, Lena Armstrong[†], F. Megumi Kivuva[†], Jayne Everson[†], and Amy J. Ko. “How Economically-Marginalized Adolescents of Color Negotiate Critical Pedagogy in a Computing Classroom” In *ACM Transactions on Computing Education (TOCE)*, 2025. **ACM TOCE Distinguished Paper.**
- 2024 Rotem Landesman^{*†}, **Jean Salac**^{*}, Jared Ordoña Lim[†], and Amy J. Ko. “Integrating Philosophy Teaching Perspectives to Foster Adolescents’ Ethical Sensemaking of Computing Technologies” In *Proceedings of the 2024 ACM Conference on International Computing Education Research (ICER)*, 2024. **Best Paper Award Honorable Mention.**
* co-first authors
Rotem Landesman[†], **Jean Salac**, Jared Ordoña Lim[†], and Amy J. Ko. “‘There Will Always be a Yes and No Side’: Facilitating Ethical Sensemaking Around Technology with Teens.” Short Paper in *International Conference of the Learning Sciences (ICLS)*, 2024.
- 2023 **Jean Salac**, Alannah Oleson, Lena Armstrong[†], Audrey Le Meur[†], and Amy J. Ko. “Funds of Knowledge used by Adolescents of Color in Scaffolded Sensemaking around Algorithmic Fairness.” In *Proceedings of the 2023 ACM Conference on International Computing Education Research (ICER)*, 2023. **Best Paper Award.**
Leah Perlmutter[†], **Jean Salac**, and Amy J. Ko. “A field where you will be accepted : Belonging in student and TA interactions in post-secondary CS education” In *Proceedings of the 2023 ACM Conference on International Computing Education Research (ICER)*, 2023.
Jean Salac, Rotem Landesman[†], Stefania Druga, and Amy J. Ko. “Scaffolding Children’s Sensemaking around Algorithmic Fairness.” In *Proceedings of the ACM Conference on Interaction Design for Children (IDC)*, 2023.
Jean Salac, Donna Eatinger, and Diana Franklin. “The Role of Spatial Orientation in Diagram Design for Computational Thinking Development in K-8 Teachers.” Research Paper in the *Proceedings of the ACM Technical Symposium on Computer Science Education (SIGCSE)*, 2023.
- 2022 Alannah Oleson^{*}, Benjamin Xie^{*}, **Jean Salac**, Jayne Everson[†], F Megumi Kivuva[†], Amy J Ko. “A Decade of Demographics in Computing Education Research : A Critical Review of Trends in Collection, Reporting, and Use” In *Proceedings of the 2022 ACM Conference on International Computing Education Research (ICER)*, 2022.
* co-first authors
- 2021 **Jean Salac**, Cathy Thomas, Chloe Butler[†], and Diana Franklin. “Investigating the Role of Cognitive Abilities in Computational Thinking for Young Learners.” In *Proceedings of the 2021 ACM Conference on International Computing Education Research (ICER)*, 2021.
Jean Salac, Cathy Thomas, Chloe Butler[†], and Diana Franklin. “Understanding the Link between Computer Science Instruction and Reading & Math Performance.” Research Paper in the *Annual Conference on Innovation and Technology in Computer Science Education (ITiCSE)*, 2021.
Jean Salac, Cathy Thomas, Chloe Butler[†], and Diana Franklin. “Supporting Diverse Learners in K-8 Computational Thinking with TIPP&SEE.” Research Paper in the *Proceedings of the ACM Technical Symposium on Computer Science Education (SIGCSE)*, 2021.
- 2020 Diana Franklin, **Jean Salac**, Zachary Crenshaw[†], Saranya Turimella[†], Zipporah Klain[†], Marco Anaya[†], Cathy Thomas. “Exploring Student Behavior Using the TIPP&SEE Learning Strategy” In *Proceedings of the 2020 ACM Conference on International Computing Education Research (ICER)*, 2020. **Best Paper Award.**
Jean Salac and Diana Franklin. “If They Build It, Will They Understand It?: Exploring the Relationship between Student Code and Performance.” Research Paper in the *Annual Conference on Innovation and Technology in Computer Science Education (ITiCSE)*, 2020.
Jean Salac, Cathy Thomas, Bryan Twarek, William Marsland, and Diana Franklin. “Comprehending Code : Understanding the Relationship between Reading and Math Proficiency, and 4th-Grade CS Learning Outcomes.” Research Paper in the *Proceedings of the ACM Technical Symposium on Computer Science Education (SIGCSE)*, 2020.
Jean Salac, Cathy Thomas, Chloe Butler[†], Ashley Sanchez[†], and Diana Franklin. “TIPP&SEE : A Learning Strategy to Guide Students through Use→Modify Scratch Activities.” Research Paper in the *Proceedings of the ACM Technical Symposium on Computer Science Education (SIGCSE)*, 2020.

- Jean Salac, Qi Jin[†], Zipporah Klain[†], Saranya Turimella[†], Max White[†], and Diana Franklin. “Patterns in Elementary-Age Student Responses to Personalized & Generic Code Comprehension Questions.” Research Paper in the *Proceedings of the ACM Technical Symposium on Computer Science Education (SIGCSE)*, 2020.
- 2019 Jean Salac, Max White[†], Ashley Wang[†], and Diana Franklin. “An Analysis through an Equity Lens of the Implementation of Computer Science in K-8 Classrooms in a Large, Urban School District.” Research Paper in the *Proceedings of the ACM Technical Symposium on Computer Science Education (SIGCSE)*, 2019.
- 2018 David Weintrop, Afsoon Afzal, Jean Salac, Patrick Francis, Boyang Li, David C. Shepherd, and Diana Franklin. “Evaluating CoBlox : A comparative study of robotics programming environments for adult novices.” In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems (SIGCHI)*, 2018. **Best Paper Honorable Mention.**
- Daniel S. Katz, Kyle E. Niemeyer, Sandra Gesing, Lorraine Hwang, Wolfgang Bangerth, Simon Hettrick, Ray Idaszak, Jean Salac, Neil Chue Hong, Santiago Nunez-Corrales, Alice Allen, R. Stuart Geiger, Jonah Miller, Emily Chen, Anshu Dubey, and Patricia Lago. “Fourth workshop on sustainable software for science : practice and experiences (WSSSPE4).” *Journal of Open Research Software* 6, no. 1 (2018).

GRANTS : OVER \$750,000 TOTAL

- 2026 Subaward Principal Investigator, FINDERS FOUNDRY Planning Proposal to National Science Foundation (\$49,942; Carleton Subaward : \$11, 517)
Title : “EdCodeGen - Reimagining Creative Computing with Pedagogical AI Code Assistance”
 Status : **Awarded** (#2627624)
- Principal Investigator, FutureCORE Proposal to National Science Foundation (Collaborative Total : \$979, 969; Carleton : \$267,975)
Title : “REPS-AI : Supporting All Students in Using GenAI Tools in Responsible, Effective, and Pedagogically Sound Ways for Computational Coursework Across Disciplines”
 Status : **Recommended for Funding**
- 2025-26 Joan Hanson Endowed Fund for Undergraduate Summer Research, Carleton College (\$16,000)
- 2022 Computing Innovations Fellowship, National Science Foundation & Computing Research Association (\$290,726)
- 2019 Graduate Research Fellowship, National Science Foundation (\$159,000)

RESEARCH STUDENTS MENTORED

- 2026-Present Matt Murashige, Undergraduate at Carleton College
- 2026-Present Zoe Voigt, Undergraduate at Carleton College
- 2025-26 Aiden Johnson, Undergraduate at Carleton College
- 2025-26 Aurelia Peterson Rajalingam, Undergraduate at Carleton College
- 2023-24 Jared Ordoña Lim, Undergraduate at University of Washington → PhD Student at Georgia Tech
- 2022-24 Rotem Landesman, PhD Student at University of Washington → Postdoc at Harvard
- 2022-23 Leah Perlmutter, PhD Student at University of Washington → Lecturer at Cornell University
- 2022-23 Lena Armstrong, Undergraduate at University of Pennsylvania → PhD Student at Harvard
- 2022 Audrey Le Meur, Undergraduate at University of Minnesota-Morris → Teacher at Teachers on Call
- 2019 Marco Anaya, Undergraduate at University of Chicago → Software Engineer at Amazon
- 2019 Zachary Crenshaw, Undergraduate at University of Chicago → Software Engineer at Epic
- 2019 Qi Jin, Undergraduate at University of Chicago
- 2019 Zipporah Klain, Undergraduate at University of Chicago → Software Engineer at Metron
- 2019 Ashley Sanchez, Undergraduate at Texas State University → Senior Technician at Dell
- 2019 Saranya Turimella, Undergraduate at University of Chicago → Associate at EY-Parthenon
- 2018-19 Chloe Butler, Masters Student at Texas State University → School Psychologist at Frisco Independent School District
- 2018-19 Max White, Undergraduate at University of Chicago
- 2018 Ashley Wang, Undergraduate at University of Chicago → Software Engineer at Meta
- 2017 Zené Sekou, Undergraduate at University of Chicago → Software Engineer at Amazon

INVITED TALKS AND WORKSHOPS

June 2026-June 2027	Professional Development Designer & Facilitator, Northern Lights Collaborative for Computing Education, University of Minnesota
May-Oct 2025	Expert Panelist, CSTA 2026 Standards for K-12 Computer Science Education, Computer Science Teachers Association
May 2025	"Emergent Gateways and Gaps in Responsible Computing Education", Department of Computer Science, Grinnell College
April 2024	"Scaffolding Youth Sensemaking around Algorithmic Fairness", Perspectives on Promoting Algorithmic Justice in Artificial Intelligence and Machine Learning Education in K-12, Symposium at American Educational Research Association
Jan-June 2024	Expert Panelist, Delphi Study for <i>Exploring an AI Literacies Framework for Young Children</i> , University at Buffalo
January 2024	"Scaffolding Children in Learning to Code with TIPP&SEE", Elementary & Middle School Computer Science Teacher Professional Development, Korea Foundation for the Advancement of Science & Creativity
November 2023	"Critical Computer Science Learning for Youth", Department of Curriculum & Instruction, University of Illinois Urbana-Champaign
November 2023	"Critical Computer Science Learning for Youth", Computer Science Research Talk, Central Michigan University
November 2023	"Critical Computer Science Learning for Youth", Computer Science Tea Talk, Carleton College
October 2023	"Critical Computer Science Learning for Youth", Computing Education Research Seminar, University of California Davis
June 2023	"Scaffolding Youth Sensemaking Around Algorithmic Fairness", Graduate Seminar, University of Pennsylvania
April 2023	"The Kids are Alright : Scaffolding Critical Computing Consciousness in Children & Adolescents", Apple Inc
April 2023	"The Kids are Alright : Scaffolding Critical Computing Consciousness in Children & Adolescents", Program in Computing for the Arts & Sciences (PCAS) Speaker Series, University of Michigan
April 2023	"The Kids are Alright : Scaffolding Critical Computing Consciousness in Children & Adolescents", School of Information Colloquium, University of Texas-Austin
February 2023	"Moving from Equity to Justice in Computing instruction for Youth", Seminar Series on Primary (K-5) Computing Education Research – Teaching and Teachers, Raspberry Pi Foundation
July 2022	"TIPP&SEE : Scaffolding for K-8 Computer Science", K-8 Research+Practice Workshop by csedresearch.org, Computer Science Teachers Association
March 2021	"Scratch Strategies to Engage Diverse Learners", Equity in Action Conference, Computer Science Teachers Association
February 2021	"Scratch Strategies to Engage Diverse Learners", Excellence in Teaching Conference, Notre Dame University
2020-21	<i>Workshop on the Next 15 Years of Computing Education Research</i> , National Science Foundation
November 2020	<i>Accessible Computer Science Education Workshop</i> , Microsoft Research

TEACHING EXPERIENCE

2024-Present	Assistant Professor Course(s) : Introduction to CS, Data Structures, Software Design, Human-Computer Interaction, (De)constructing Everyday Technologies, Senior Capstone Carleton College
2023	Co-Instructor Course(s) : Philosophy behind Everyday Technologies University of Washington Upward Bound Summer Academy
2022	Lead Instructor Course(s) : Foundations of Computing with a Critical Lens University of Washington Upward Bound Summer Academy
2017-19	Teaching Assistant Course(s) : Computers for Learning (Game Engine Design, Introduction to Object-Oriented Programming) University of Chicago
2016-17	Teaching Assistant Course(s) : Human-Computer Interaction & Computer Architecture University of Virginia
2013-17	Museum Educator National Air & Space Museum

SELECT SERVICE

- 2026-2027 Guest Editor, Transactions on Computing Education on Special Issue “Situating and Critical Effects of Generative AI in Computing Education : Identity, Belonging, and the Social Conditions of Learning”
- 2025-Present Higher Education Representative, Computer Science Teachers Association - Minnesota Chapter (CSTA-MN)
- 2021-Present Program Committee Member, International Computing Education Research (ICER) Conference
- 2021-2023 Social Media Coordinator, International Computing Education Research (ICER) Conference
- 2021-Present Peer Reviewer, CHI, IDC, Communications of the ACM (CACM), Journal of Computer Science Education (CSE), Transactions of Computing Education (TOCE), Transactions of Human Factors in Computing (ToCHI), Transactions of Human-Robot Interaction (ToHRI)
- 2023 Program Committee Member, Conference on Primary & Secondary Computing Education Research (WiPSCE)
- 2023 Proposal Review Panelist, NSF Computer and Information Science and Engineering (CISE)
- 2018-21 Computer Science Representative, University of Chicago : (1) Graduate Recruitment Initiative Team & (2) Physical Sciences Division’s Committee on Equity, Diversity, & Inclusion
- 2018-20 Co-Organizer, University of Chicago Research Symposia : (1) Women in STEM Symposium & (2) Transcending Boundaries Research Symposium for Scholars of Color
- 2018-19 Co-Chair & Founder, University of Chicago Graduate Women in Computer Science